Microbiome
Normal Flora of Human Body
Gut Microbiome

Normal Flora

Normal flora is the term used to describe the various bacteria and fungi that are permanent residents of certain body sites, especially the skin, oropharynx, colon, and vagina.

Viruses and parasites are never considered members of the normal flora.

Normal flora are (commensalism): organisms that derive benefit from another host but do not damage that host.

Resident flora

– **Species that live on or in nearly everyone almost all the time.**

– **These residents live in specific sites**

- ***Staph. Epidermidis*** **(skin)**
- ***E. coli*** **(colon and small intestine)**

– **They are non-pathogenic and may become pathogenic if introduced into abnormal sites**

Transient flora

- **Species that are found periodically on or in the body**

  - Example: *Streptococcus pneumoniae*

- **Transients may become pathogenic when the host's resistance is lower.**

- ***Colonization* refers to the acquisition of a new organism (mainly pathogenic, on mucosal surfaces).**

THE HUMAN MICROBIOME:

- **Human microbiome:** All microorganisms (bacteria, fungi, and viruses) that inhabit the human body.

- The largest and complex microbial population resides in the colon.

- The intestinal microbiota also plays a role in certain autoimmune diseases such as the inflammatory bowel diseases

- The gut microbiota influences the maturation and function of the immune response and colonization resistance .

Normal flora play a role both in the maintenance of health and in the causation of disease in three significant ways:

1 - Their presence in immunocompromised individuals.

2- They constitute a protective host defense mechanism.

They occupy attachment sites on the skin and mucosa that can interfere with colonization by pathogenic bacteria. (colonization resistance).

3- Nutritional function. The intestinal bacteria produce vitamin B and vitamin K. (treated with oral antibiotics leads to vitamin deficiencies).

Composition and Development of the Gut Microbiome

- Gut microbial colonization develops early in life, (influenced by birth and environmental exposures).

- The neonatal gut microbiome has relatively low diversity compared with the adult microbiome.
- Adult gut microbiomes may contain more than one thousand distinct microbial species.

Factors Affecting Gut Microbiome

- Differences in oxygen, pH, nutrients, and mucus influence microbial distribution throughout the gastrointestinal tract.

- Diet, genetics, medications, household contacts, and environmental exposures contribute to individual microbiome variation.

- **Antibiotics can disrupt microbial communities**, eliminating susceptible organisms and allowing others to proliferate.

Advantages of gut microbiome

- Host nutrition

- Prevents colonization
  - Competing for receptors or binding sites
  - Competing for nutrients
  - Producing antibiotics or bacteriocins
  - Production of toxic products

- Stimulation of natural antibodies that cross react with pathogenic organisms

Metabolic Functions of the Gut Microbiome

Gut microbes ferment dietary fibers and complex carbohydrates that human digestive enzymes cannot fully break down.

This fermentation produces short-chain fatty acids, including acetate, propionate, and butyrate, that support intestinal barrier integrity, immune regulation.

Gut microorganisms also synthesize essential vitamins, including vitamin K and several B vitamins.

Disadvantages of normal flora:

- They become pathogenic when the host resistance is lowered
- They may act as pathogens in tissue outside their normal habitat

Examples on normal flora disadvantages:

- Intestinal flora may cause urinary tract infections (UTI)

- *Bacteroides fragilis*: cause of peritonitis associated with perforation of intestinal wall following trauma, appendicitis or diverticulitis

- *Streptococci viridans (S.mutans) could be* introduced into the bloodstream following tooth extraction or tonsillectomy and cause (Infective endocarditis).

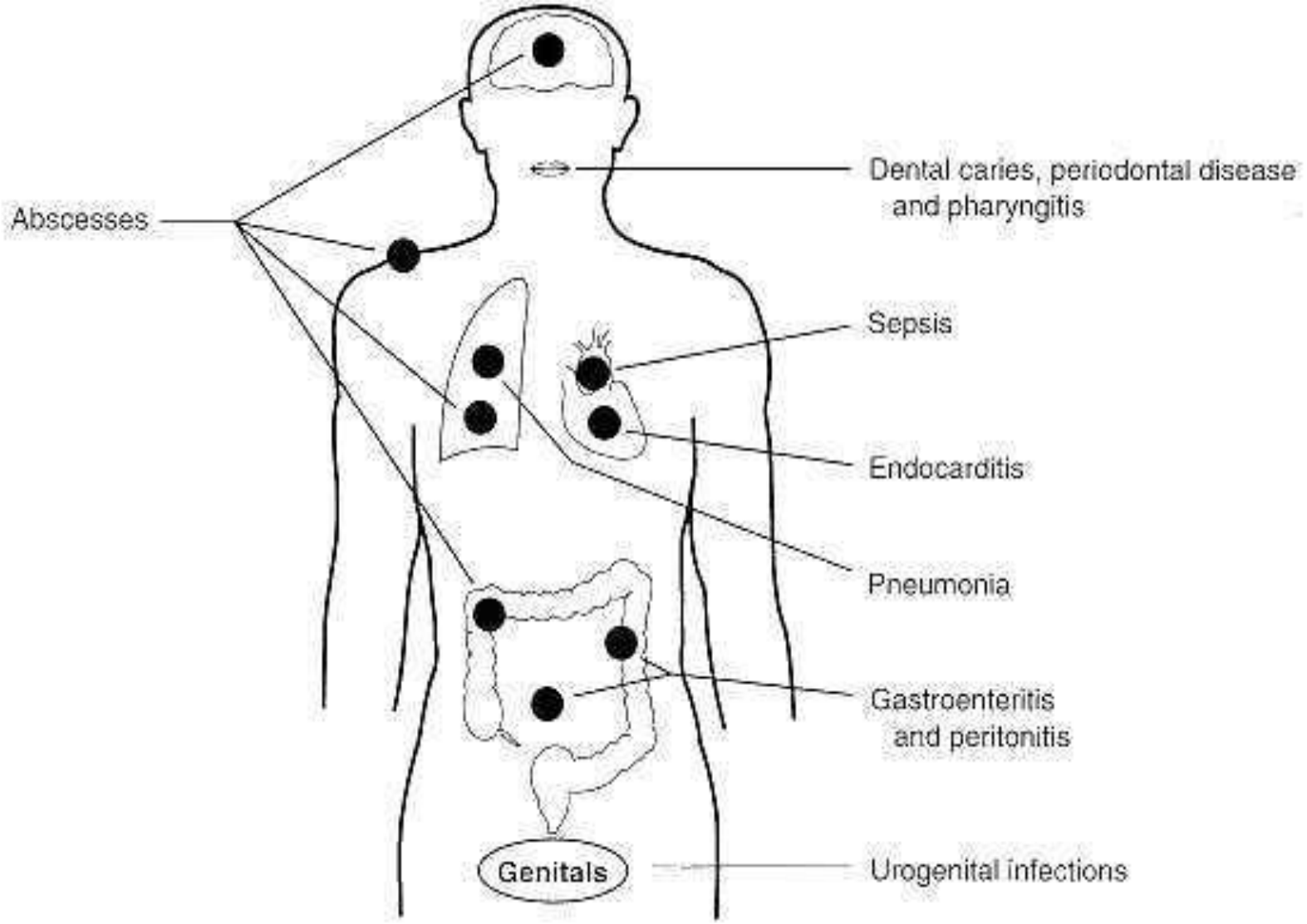
The Gut Microbiome in Systemic Disease and Cancer

Gut microbiome alterations are associated with type 2 diabetes and inflammatory bowel diseases.

Microbial changes may influence pathways involved in insulin resistance, intestinal permeability, inflammation,

Helicobacter pylori **can contribute to gastric adenocarcinoma and mucosa-associated lymphoid tissue lymphoma.**

Other bacteria, including *Fusobacterium nucleatum*, are implicated in colorectal cancer development.


Abscesses
Dental caries, periodontal disease and pharyngitis
Sepsis
Endocarditis
Pneumonia
Gastroenteritis and peritonitis
Genitals
Urogenital infections

Effect of Antibiotics on Normal Flora

- Resident flora may be diminished by the use of antibiotics to treat bacterial infections
- An antibiotic does not distinguish between the pathogen and the resident bacteria
- Without the usual competition, yeasts or pathogenic bacteria may be able to overgrow and create new infections.

Examples on effect of antibiotics on intestinal normal flora

- **Clindamycin can suppress the predominant normal flora allowing a rare toxin-producing *Clostridium difficile* to overgrow and cause severe colitis**

- **Neomycin given prior to GI surgery to sterilize gut leads to decrease bacterial levels but it return to normal after several days**

Localizations of the Normal Flora

- Skin
- Eyes (conjunctiva)
- Nose and throat (upper respiratory tract)
- Mouth
- Large Intestine
- Urinary and reproductive systems

TABLE 6–1 Summary of the Members of Normal Flora and Their Anatomic Locations

Members of the Normal Flora[1]	Anatomic Location
Bacteroides species	Colon, throat, vagina
Candida albicans	Mouth, colon, vagina
Clostridium species	Colon
Corynebacterium species (diphtheroids)	Nasopharynx, skin, vagina
Enterococcus faecalis	Colon
Escherichia coli and other coliforms	Colon, vagina, outer urethra
Gardnerella vaginalis	Vagina
Haemophilus species	Nasopharynx
Lactobacillus species	Mouth, colon, vagina
Neisseria species	Mouth, nasopharynx
Propionibacterium acnes	Skin
Pseudomonas aeruginosa	Colon, skin
Staphylococcus aureus	Nose, skin
Staphylococcus epidermidis	Skin, nose, mouth, vagina, urethra
Viridans streptococci	Mouth, nasopharynx

[1]In alphabetical order.

NORMAL FLORA OF THE SKIN

The predominant organism is *Staphylococcus epidermidis*, which is a nonpathogen on the skin but can cause disease when it reaches certain sites, such as artificial heart valves and prosthetic joints.

Anaerobic organisms, such as *Propionibacterium* and *Peptococcus (to delete)*, are situated in the deeper follicles in the dermis. *Propionibacterium acnes* is a common skin anaerobe that is implicated in the pathogenesis of acne.

The yeast *Candida albicans* is also a member of the normal flora of the skin.

New name for *Propionibacterium acnes* is *Cutibacterium acnes*

NORMAL FLORA OF THE RESPIRATORY TRACT

They colonize the nose, throat, and mouth (mainly streptococcal and staphylococcal), but not in the lower bronchi and alveoli.

The throat contains a mixture of viridans streptococci, *Neisseria* species, and *S. epidermidis* . These nonpathogens inhibit the growth of the pathogens *Streptococcus pyogenes, Neisseria meningitidis,* and *S. aureus,* respectively.

Viridans group in mouth (green colonies):

***Streptococcus mutans*, a member of the viridans group, found in large numbers in dental plaque, the precursor of caries.**

The viridans streptococci, are also cause subacute bacterial (infective) endocarditis.

NORMAL FLORA OF THE INTESTINAL TRACT:

The small intestine usually contains small numbers of streptococci, lactobacilli, and yeasts, particularly *C. albicans*.

The colon is the major location of bacteria in the body. Roughly 20% of the feces consists of bacteria, approximately 10^{11} organisms/g.

More than 90% of the fecal flora are anaerobes, the most important of which is *Bacteroides fragilis*.

The most abundant facultative bacteria are the coliforms, (*Escherichia coli*).

The normal flora of the intestinal tract plays a significant role in extra-intestinal disease. For example, *E. coli* is the leading cause of urinary tract infections, and *B. fragilis* is an important cause of peritonitis associated with perforation of the intestinal wall following trauma, appendicitis, or diverticulitis (inflammation in the large intestine).

Other important pathogens that could cause UTI and endocarditis:

- *Fusobacterium* and *Peptostreptococcus (*anaerobic).
- *Enterococcus faecalis,* (facultative bacteria) cause endocarditis.
- *Pseudomonas aeruginosa,* causing infection in hospitals (*It* is present in 10% of normal stools, as well as in soil and water).

NORMAL FLORA OF THE GENITOURINARY TRACT

Lactobacillus species are normal flora of the vagina (in adults) that produce acidic pH.

Lactobacilli could prevent the growth of potential pathogens, since antibiotics treatment can lead to overgrowth by *C. albicans.*

During urine passages through urethra it becomes contaminated with *S. epidermidis,* coliforms, diphtheroids, and nonhemolytic streptococci.

Mycobacterium smegmatis: mainly found in the secretions around the urethral areas.
Staphylococcus saprophyticus: found on the skin surrounding the genitourinary tract and it could be a cause of UTI in women.

Summary of diseases caused by normal flora

- *E. Coli* → UTI
- *Bacteroides fragilis* → Peritonitis (perforation of intestinal wall following trauma, appendicitis)
- *Enterococcus feacalis* → UTI & endocarditis
- *Pseudomonas aeruginosa* → various infections in patients with low host defenses

Diseases caused by NF of Genitourinary tract

- **Suppression of Lactobacilli by antibiotics will cause overgrowth of the yeast-like fungi:**
- ***Candida albicans*** → **"*Candida* vaginitis"**

- **Recurrent UTI s due to *E. Coli* and *Enterobacter***

- **15-20% women of childbearing age carry group B streptococci (GBS) in the vagina→ (sepsis and meningitis in the newborn)**

Probiotics

- **The use of normal flora of the GI tract to treat diseases**
- **It stabilizes the composition of the Normal flora**
- **10^9-10^{10} organisms: Is the quantity of microorganisms used in some probiotic preparations**

Clinical Applications and Microbiome-Based Therapies

- **Probiotics:** are live microorganisms that provide health benefits to our body when consumed in adequate amounts.

- The use of normal flora of the GI tract to treat diseases

- **Fecal microbiota transplantation** restores microbial communities and effectively treats recurrent *Clostridioides difficile* infection.

- Microbiome-based interventions are being investigated for inflammatory bowel disease and other conditions.

Probiotics

- **Probiotic bacteria should have the following characteristics:**

  - Beneficial physiological effect
  - Strain is of human origin
  - Safe for human use
  - Stable in acid and bile
  - Adhere to intestinal mucosa
  - Most common:

    - Lactobacilus acidophilus
    - Bifidobacterium bifidum
    - Streptococcus thermophilus

Use of Probiotics in treatment of diseases

1- Treatment and prevention of diarrhea

- Rotavirus and Travelers' diarrhea in children
- Post-antibiotic diarrhea long-term use of antibiotics

2- Alleviate symptoms of lactose intolerance: (lactic acid bacteria)

3- Food allergies: e.g. milk.

4- Treatment of GI inflammations associated with disruption of the mucosal barrier/permeability:

- Crohn's disease
- Food allergies
- Atopic eczema